Conclusion

EDA Analysis – Credit Card Customer Dataset

- The dataset provides insights into customer spending behavior, credit usage, payments, and financial patterns.
- Most customers actively use their credit cards for **purchases, cash advances, and installment payments**.
- Customers with **higher credit limits** generally show **higher purchase frequency and transaction amounts**.
- A strong relationship is observed between **purchase amount and payment amount**, meaning customers who spend more also tend to repay more.
- Some customers maintain **high balances with low payments**, which may indicate a risk of delayed payments or debt accumulation.
- Cash advance usage is comparatively lower than regular purchases, showing that customers prefer direct card spending instead of withdrawing cash.
- Several variables contain **skewed distributions and outliers**, especially in balance, purchases, and cash advance amounts, indicating different customer spending behaviors.
- Frequent purchasers usually have **better payment consistency**, making them more reliable customers for the bank.
- Customers with low transaction frequency may represent inactive users or low-engagement customers.
- Overall, the analysis shows that customer behavior is mainly influenced by **credit limit, spending frequency, repayment habits, and balance management**.

Recommendations

- The bank should target **high-value customers** with premium offers, reward programs, and higher credit benefits.
- Customers with **high balances and low payments** should be monitored closely to reduce default risk.
- Personalized offers and cashback programs can help increase engagement among low-frequency users.
- Encourage customers to use cards for regular purchases instead of cash advances through reward-based incentives.
- Introduce financial awareness campaigns for customers with poor repayment behavior to improve payment consistency.
- Outlier customers with extremely high spending can be segmented separately for VIP services and risk assessment.
- Predictive models can be developed using this data to identify potential defaulters and loyal customers in advance.